Assignment: Why do investors stop their SIPs?

25 business scenarios, from cleaning the data to predicting with K-Nearest Neighbors (KNN)

The business situation

Sahyadri Mutual Fund is a fictional mutual fund company. Most of its investors invest through a **SIP** (Systematic Investment Plan): a fixed amount is taken from the investor's bank account every month and invested in a fund. The company earns a small yearly fee on the money it manages, so an investor who keeps the SIP running for years is valuable. An investor who stops the SIP is a loss.

On 31 March 2026 the company took a snapshot of about 2,000 SIP investors. Six months later it recorded which of them had **stopped** their SIP. The head of retention, Mr. Kulkarni, wants two things from you:

1. **Find out why** investors stop their SIPs.
2. **Predict which investors are likely to stop**, so that the retention team can call them before they do.

You will work through 25 scenarios. Each one is a question asked by someone in the company. You answer it with code, a table or chart, and two or three sentences in plain language.

Files you receive

`sip_investors.csv`: one row per investor, 13 columns. The file comes straight from the company's systems and has **not** been cleaned.

The columns

Column	Meaning
`investor_id`	Unique ID of the investor. One investor should appear only once.
`age`	Age of the investor in years.
`city_tier`	Size of the investor's city: Metro, Tier-2 or Tier-3.
`channel`	How the investor was signed up: Direct App (the company's own app), Distributor (an independent agent), or Bank RM (a relationship manager at a partner bank).
`fund_category`	Type of fund the SIP goes into: Large Cap, Flexi Cap, Mid Cap, Small Cap or ELSS. Small Cap and Mid Cap funds go up and down in value more than Large Cap funds.
`monthly_income`	Investor's monthly income in rupees, as declared at sign-up.
`sip_amount`	Amount invested every month through the SIP, in rupees.
`sip_tenure_months`	Number of months since the SIP started, on the snapshot date.
`missed_sips_12m`	Number of monthly instalments that failed in the last 12 months (for example, because the bank account did not have enough money).
`portfolio_return_pct`	Gain or loss on the investor's money so far, in percent. +10 means the investment is worth 10% more than what was put in. -10 means 10% less.
`app_logins_per_month`	Average number of times per month the investor opened the app to look at the investment.
`num_funds_held`	Number of different funds the investor holds with the company.
`sip_stopped`	**The answer (target).** 1 = the investor stopped the SIP in the six months after the snapshot. 0 = the SIP is still running.

Rules that keep everyone's numbers the same

Follow these exactly. Your results will be compared with a reference solution that uses the same settings.
• Use Python with pandas, numpy, matplotlib, seaborn and scikit-learn. Solve the scenarios **in order**. Each scenario continues from the data left by the previous one.
• Wherever a random choice is made, use `random_state=42`.
• Train/test split: `test_size=0.2`, with `stratify` on `sip_stopped`.
• Cross-validation: always `cv=5`, always on the training set only.
• Outlier rule: the IQR rule with fences at Q1 − 1.5 × IQR and Q3 + 1.5 × IQR.
• Any value learned from data (a median, a fence, a scaling average) is learned from the **training set** and then applied to both the training set and the test set.
• Never use the test set to make a choice (columns, K, threshold). Test results appear only in Scenarios 23 to 25.

Before you start (not marked)

Load the file into a DataFrame. Look at the first rows (`head`), the column types and non-empty counts (`info`) and the summary numbers (`describe`). Write down three things in these outputs that look wrong or suspicious. You will meet each of them again in Part A.

Part A: Clean the data (Scenarios 1 to 5)

Scenario 1: The same investor, twice

4 marks

Farah, data engineer: *"Our nightly export sometimes writes a record twice when the connection drops. Is any investor repeated in your file?"*

(a) Count the rows whose `investor_id` has already appeared earlier in the file.

(b) Show all rows for one repeated investor, to confirm that the rows are identical.

(c) Keep the first copy of each investor and remove the rest.

Write in your notebook: the number of repeated rows, and the number of rows before and after. Explain in one line why a repeated row would make the company's stop rate wrong.

Scenario 2: Twelve city tiers

4 marks

Rohit, regional sales head: *"My report by city tier shows twelve tiers. We have three. I cannot compare regions like this."*

(a) List every distinct value in `city_tier` with its count.

(b) Clean the column so that it holds exactly three values: `Metro`, `Tier-2` and `Tier-3`.

(c) Show the counts again after each cleaning step, not only at the end.

Write in your notebook: the counts before and after. Name one value that looks identical to another on screen but is different to the computer, and one value that no automatic rule could fix.

Scenario 3: An investor aged 199

4 marks

Neha, compliance officer: *"Your summary shows an investor aged 199, an investor aged 0, and a return of minus 999 percent. Are these real?"*

(a) Use these valid ranges: `age` from 18 to 100, `portfolio_return_pct` from −100 to +200.

(b) For each of the two columns, count the values outside the range and show which values they are.

(c) Replace only those values with an empty cell (`NaN`). Do not delete the rows.

(d) Show the average of both columns before and after the fix.

Write in your notebook: what these values most likely are, how much they had moved the two averages, and why you kept the rows.

Scenario 4: Whose income is missing?

4 marks

Neha, compliance officer: *"Income is a required field on the sign-up form. I want to know how often it is empty, and who is leaving it empty."*

(a) Count the empty cells in every column, as a number and as a percentage of rows.

(b) For the column with the most empty cells, calculate the percentage of empty cells for each `channel`.

(c) Draw this as a bar chart.

Write in your notebook: which column is worst, which channel is responsible, and one action Neha should take.

Scenario 5: How big is the problem, in rupees?

4 marks

Mr. Kulkarni, head of retention: *"Before I put a team on this, tell me how much money we are losing."*

(a) Count the investors who stopped, and calculate the **stop rate** (the percentage who stopped).

(b) Finance says each stopped SIP costs the company ₹1,000 in lost future fees. Calculate the loss for the investors in this file.

(c) The company has 4,00,000 SIP investors. Estimate the loss for all of them, assuming the same stop rate.

(d) Draw a bar chart of the number of investors who continued and who stopped.

Write in your notebook: the stop rate, both rupee figures, and one more number: the accuracy of a model that says "will continue" for every investor without looking at anything. Any model you build later has to beat it.

Part B: Find the causes (Scenarios 6 to 12)

Several scenarios in this part ask the same kind of question: "what is the stop rate for each value of this column?" Write one small function that takes a column name and returns the stop rate (%) and the number of investors for each value, and reuse it. In every bar chart, draw the company's overall stop rate as a dashed horizontal line.

Scenario 6: Which channel loses the most investors?

4 marks

Rohit, regional sales head: *"We pay commission to distributors and to bank relationship managers. Are the investors they bring as loyal as the ones who sign up on our own app? And does the size of the city matter?"*

(a) Calculate the stop rate and the number of investors for each `channel`. Draw a bar chart.

(b) Do the same for `city_tier`.

Write in your notebook: which channel is worst and by how many percentage points, whether city tier makes a difference, and what Rohit should look into.

Scenario 7: Should we stop selling Small Cap SIPs?

4 marks

Anita, product head: *"I hear that Small Cap investors leave far more often than the others. If that is true, I will stop promoting Small Cap SIPs."*

(a) Calculate the stop rate for each `fund_category` and draw a bar chart. Note what Anita would conclude from this chart alone.

(b) Show the average, lowest and highest `portfolio_return_pct` for each fund category.

(c) Cut `portfolio_return_pct` into five bands: below −10, −10 to 0, 0 to 10, 10 to 20, above 20.

(d) Build a grid with fund categories as rows and return bands as columns. Each cell is the stop rate for that combination. Draw it as a heatmap.

(e) Build a second grid showing the number of investors in each cell.

Write in your notebook: read the heatmap one column at a time. Among investors with the **same level of return**, is Small Cap clearly worse than the other categories? What is the real reason Small Cap looked bad in (a)? Which cells have too few investors to trust? What should Anita do?

Scenario 8: When do we lose them?

4 marks

Mr. Kulkarni, head of retention: *"I can afford a welcome programme with reminder messages and one call from an adviser. In which months of an investor's journey should it run?"*

(a) Cut `sip_tenure_months` into bands: 1–6, 7–12, 13–18, 19–24, 25–36.

(b) Calculate the stop rate and number of investors for each band. Draw a bar chart.

Write in your notebook: in which period the risk is highest, what happens to the risk after that, and when the welcome programme should run.

Scenario 9: Is a missed instalment a warning sign?

4 marks

Imran, payments operations: *"When a monthly instalment fails, we only send an automatic SMS. Should a failed instalment trigger something more?"*

(a) Calculate the stop rate and number of investors for each value of `missed_sips_12m`. Draw a bar chart.

Write in your notebook: how the stop rate changes with each additional missed instalment, after how many misses an alert to the retention team makes sense, and which bars rest on too few investors to be trusted.

Scenario 10: "Investors in profit are safe"

4 marks

Mr. Kulkarni, head of retention: *"My team calls only the investors who are sitting on losses. Investors in profit are happy, so we leave them alone. Is that right?"*

(a) Cut `portfolio_return_pct` into six bands: below −15, −15 to −5, −5 to 5, 5 to 15, 15 to 25, above 25.

(b) Calculate the stop rate and number of investors for each band. Draw a bar chart.

Write in your notebook: the shape of the chart, the return band with the lowest stop rate, and the **two** groups of investors who stop most often. Give one plausible reason for each group. Is Mr. Kulkarni's rule right?

Scenario 11: Is the SIP too big for the salary?

4 marks

Priya, financial planner: *"I think people stop because they signed up for more than they can afford. But a ₹10,000 SIP is small for one person and large for another. Can you check this properly?"*

(a) Create a new column: `sip_to_income_pct = sip_amount / monthly_income * 100`. It shows what share of the monthly income goes into the SIP.

(b) Cut it into bands: below 5, 5–10, 10–15, 15–20, above 20. Calculate the stop rate for each band and draw a bar chart.

(c) For comparison, split `sip_amount` into four groups of equal size (`pd.qcut`) and show the stop rate of each group. Do the same for `monthly_income`.

Write in your notebook: which tells you more about stopping: the two original columns, or the new one made from them? Above what percentage of income should the app show a warning at sign-up?

Scenario 12: "Correlation says it does not matter"

4 marks

Varun, junior analyst: *"I calculated the correlation of every column with sip_stopped. Returns come out weak and negative, so the more an investor gains, the safer they are. Several other columns are close to zero, so I will drop them."*

(a) Calculate the correlation of each numeric column (including `sip_to_income_pct`) with `sip_stopped`. Draw a horizontal bar chart.

(b) Create a temporary value: how far the return is from 8%, in either direction (`abs(portfolio_return_pct - 8)`). Calculate its correlation with `sip_stopped`.

(c) For `age`, `num_funds_held` and `monthly_income`: split each into four equal-sized groups and show the stop rate of each group.

Write in your notebook: compare what correlation says about returns with what you saw in Scenario 10. Why does correlation understate the effect of returns? Which columns really show no link with stopping, and what is your evidence beyond the correlation number?

Part C: Outliers (Scenarios 13 and 14)

An outlier is a value that sits far away from most other values in its column. In Scenario 3 you removed values that were **impossible**. What is left now are values that are possible but unusual.

Scenario 13: Which values are unusual?

4 marks

Neha, compliance officer: *"Give me a list of the columns where some investors are far outside the normal range."*

(a) Draw a box plot for each of the nine numeric columns (the eight original ones and `sip_to_income_pct`).

(b) For `app_logins_per_month`, work out the IQR rule step by step and print each value: Q1, Q3, IQR, lower fence, upper fence, number of investors beyond each fence, and the largest value.

(c) Write a function that returns the two fences for a column. Use it to build one table for all nine columns: lower fence, upper fence, smallest value, largest value, number of outliers, percentage of outliers.

Write in your notebook: which columns have more than 3% outliers, and for each of them, whether the outliers are on the high side, the low side or both.

Scenario 14: "Just delete the outliers" — 4 marks

Varun, junior analyst: *"Outliers spoil the averages. The simplest fix is to delete every investor who has one."*

The business teams confirmed that these values are real. Logins above about 40 a month come from investors who have linked a portfolio-tracking app that refreshes automatically. Very high incomes and SIP amounts belong to wealthy investors.

(a) Study the upper outliers of these five columns: `app_logins_per_month`, `monthly_income`, `sip_amount`, `missed_sips_12m`, `sip_to_income_pct`.

(b) For each column, split the investors into three groups: normal (not above the upper fence), outliers in the nearer half, and outliers in the farther half. Use the median of the outlier values as the dividing point. Calculate the stop rate of each group.

(c) Count the investors who have an outlier in at least one of the nine numeric columns, and how many of them stopped.

Write in your notebook: for each of the five columns, answer two questions. Do the outlier investors behave differently from normal ones? Does the **size** of the outlier matter (is the farther half clearly different from the nearer half)? Then decide for each column: delete, cap, or leave as it is, with one line of reasoning. Finally, state what deleting all outlier investors would cost.

Part D: Prepare the data for the model (Scenarios 15 to 19)

The steps happen in this order, and the order matters: split → fill missing values → cap outliers → convert text to numbers → scale.

Scenario 15: Prove it on investors it has not seen — 4 marks

Ms. Rao, internal auditor: *"Any model looks good on the data it was built from. How will you show me that it works on new investors?"*

(a) Split the cleaned table into a training set (80%) and a test set (20%), using the settings in the rules box.

(b) Print the number of investors and the stop rate in each set.

(c) Calculate the do-nothing accuracy on the test set (predict "will continue" for everyone).

Write in your notebook: the sizes, the two stop rates, the do-nothing accuracy, and one or two lines on why the split is done before filling, capping and scaling.

Scenario 16: Score the investor even when a value is missing — 4 marks

Rohit, regional sales head: *"Many distributor forms come in without income. Your model cannot simply refuse to score those investors."*

(a) Calculate the median of each of the nine numeric columns from the training set.

(b) Fill the empty cells in both sets with these training medians. Confirm that no empty cells are left.

(c) Print the average and the median of `monthly_income` in the training set.

Write in your notebook: why the median is a safer filling value than the average for this data, with the two income figures as evidence.

Scenario 17: Keep the investor, remove the distortion — 4 marks

Neha, compliance officer: *"We agreed not to delete anyone. So what do we do with the extreme values?"*

Capping means: a value above the upper fence is replaced by the upper fence, and a value below the lower fence is replaced by the lower fence.

(a) Keep a copy of the training and test sets as they are now (filled, not capped). You need it in Scenario 20.

(b) Cap these three columns at their IQR fences, learned from the training set: `app_logins_per_month`, `monthly_income`, `sip_amount`. Leave `missed_sips_12m`, `sip_to_income_pct` and `portfolio_return_pct` as they are.

(c) For each capped column, print before and after: largest value, average, standard deviation, number of values changed, and correlation with `sip_stopped`.

(d) Draw box plots of `app_logins_per_month` before and after capping.

Write in your notebook: what happened to the correlation between logins and stopping after capping, and why it changed even though no information was added. If your decision in Scenario 14 was different from the list in (b), say so in one line, but use the list in (b) so that your later numbers match.

Scenario 18: The model cannot read "Bank RM" — 4 marks

Rohit, regional sales head: *"The channel clearly matters. But you told me KNN works only with numbers."*

(a) Convert `city_tier`, `channel` and `fund_category` into 0/1 columns with `pd.get_dummies(..., dtype=int)`.

(b) Make sure the test set has the same columns in the same order as the training set.

(c) Build the input tables: the nine numeric columns plus the new 0/1 columns. Separate the target `sip_stopped` for each set.

Write in your notebook: the number of input columns, and why numbering the channels 1, 2 and 3 would have been wrong.

Scenario 19: Income is in lakhs, missed instalments go from 0 to 6

4 marks

Priya, financial planner: *"You compare investors by subtracting their values. Won't income swamp everything else?"*

(a) Fit a `StandardScaler` on the training inputs. For `monthly_income`, `sip_amount`, `missed_sips_12m` and `portfolio_return_pct`, show the smallest and largest value before and after scaling.

(b) Using 5-fold cross-validation on the training set, measure the accuracy of KNN with K = 5 on all input columns, once without scaling and once with scaling. For the scaled version use `make_pipeline(StandardScaler(), KNeighborsClassifier(5))`.

Write in your notebook: three numbers side by side: do-nothing accuracy on the training set, KNN without scaling, KNN with scaling. Explain the result.

Part E: Build the KNN model and put it to work (Scenarios 20 to 25)

Scenario 20: "We have twenty columns. Use them all."

4 marks

Mr. Kulkarni, head of retention: *"More information should give better predictions. Why would you leave anything out?"*

(a) From your findings in Part B, list every original column with "keep" or "drop" and the scenario that supports your decision.

(b) Compare two sets of columns with 5-fold cross-validation (K = 5): all input columns, and these six: `missed_sips_12m`, `sip_to_income_pct`, `sip_tenure_months`, `portfolio_return_pct`, `app_logins_per_month`, `channel_Bank RM`. Measure each set with and without scaling, and show the four accuracies in one table.

(c) Check whether capping helped. Using the six columns with scaling, run `cross_val_predict` twice: once with the capped `app_logins_per_month` and once with the uncapped copy from Scenario 17. For each, report accuracy, stoppers caught and false alarms.

Write in your notebook: why fewer columns can give a better KNN model, and whether capping earned its place. Use these six columns for the remaining scenarios.

Scenario 21: Show me one prediction, step by step

4 marks

Priya, financial planner: *"Before I trust this, walk me through one investor. No library, just the steps."*

(a) Take the first investor in your test set. Scale the six columns using the training averages and standard deviations.

(b) With numpy only, calculate the Euclidean distance from this investor to every training investor.

(c) Show the 5 nearest training investors: their six values, their distance, and whether they stopped.

(d) Take the majority vote. Compare your prediction with what really happened and with the prediction of `KNeighborsClassifier(5)`.

Write in your notebook: the vote count, the prediction, and whether the five neighbours really look similar to the investor.

Scenario 22: How many neighbours should vote?

4 marks

Varun, junior analyst: *"You used K = 5 everywhere without giving a reason."*

(a) For every odd K from 1 to 45, record the training accuracy and the 5-fold cross-validation accuracy (scaled pipeline, six columns).

(b) Draw both as lines against K, with the do-nothing accuracy as a horizontal line. Mark the K with the highest cross-validation accuracy.

Write in your notebook: the best K. Explain the training accuracy at K = 1. Describe what happens to both lines as K grows, and why.

Scenario 23: Open the locked box

4 marks

Ms. Rao, internal auditor: *"You have finished tuning. Show me the result on the test investors."*

(a) Fit the pipeline with your best K on the full training set. Predict the test set.

(b) Print the accuracy next to the do-nothing accuracy.

(c) Draw the confusion matrix as a heatmap with clear row and column labels, and print the classification report.

Write in your notebook: each of the four cells of the confusion matrix in words. Calculate by hand, from the cells, the share of real stoppers the model caught (recall) and the share of flagged investors who really stopped (precision). Is accuracy a good way to judge this model? Why?

Scenario 24: Who should the retention team call?

4 marks

Mr. Kulkarni, head of retention: *"A retention call costs me ₹250. A stopper I did not call costs me ₹1,000. I would rather make a few unnecessary calls than miss a stopper. Can the model work that way?"*

Assume that a call always persuades an investor who was about to stop. So an investor who is called costs ₹250 whatever happens, and a stopper who is not called costs ₹1,000.

(a) On the test set, calculate the cost per 1,000 investors of three options: call nobody, call everybody, and call only the investors the Scenario 23 model predicts as stoppers.

(b) Work out by hand: above what chance of stopping is a call worth its cost?

(c) On the training set, use `cross_val_predict(..., method="predict_proba")` to get, for each investor, the share of neighbours who stopped. Do this for K = 5, 9, 15, 21, 25, 31 and 41. For each K, try every rule of the form "call when j or more of the K neighbours stopped" and calculate the cost per 1,000 investors.

(d) Pick the K and j with the lowest cost. Draw cost against threshold (j ÷ K, in %) for K = 5 and for your best K on one chart.

(e) Apply the chosen rule to the test set. Report investors called, stoppers caught, stoppers missed, unnecessary calls, and cost per 1,000 investors.

Write in your notebook: the chosen rule in one sentence. Compare its threshold with your answer to (b). Compare its cost with the three options in (a), and estimate the saving for 4,00,000 investors against calling nobody. Why is the best K here different from the best K in Scenario 22?

Scenario 25: This week's call list, and the final report

4 marks

Mr. Kulkarni, head of retention: *"Here are five investors from this week. Tell me who to call. Then give me your findings on one page."*

(a) Prepare the five investors below exactly as you prepared the training data: create `sip_to_income_pct`, cap `app_logins_per_month` with the training fence, and create `channel_Bank RM`.

(b) Apply the rule from Scenario 24. For each investor show the number of neighbours who stopped, the share, and the decision (call or do not call).

(c) For each investor, also show the average distance to the neighbours (`kneighbors` on the fitted model).

(d) For the investor with the highest share, list the first ten neighbours with their six values and whether they stopped.

Write in your notebook: which investors to call. Which one prediction do you trust least, and why? Use the average distance and what you learned in Scenario 10. Then finish with a summary table for Mr. Kulkarni with at least six rows: **finding, evidence (with numbers and scenario), recommended action**.

Column	Investor A	Investor B	Investor C	Investor D	Investor E
`missed_sips_12m`	0	3	1	0	0
`sip_amount`	5,000	15,000	6,000	8,000	10,000
`monthly_income`	90,000	60,000	70,000	1,20,000	2,50,000
`sip_tenure_months`	24	4	3	30	14
`portfolio_return_pct`	9.0	−12.0	2.0	28.0	10.0
`app_logins_per_month`	8	1	4	10	150
`channel`	Direct App	Bank RM	Distributor	Direct App	Distributor

Check your numbers

These checkpoints do not give away any finding. If one of them does not match, fix it before moving on, because every later scenario depends on it.

After scenario	You should have
1	2,000 rows
2	Exactly three values in `city_tier`
3	12 ages and 14 returns replaced with empty cells. No rows deleted.
15	1,600 training investors and 400 test investors
18	20 input columns (9 numeric + 11 new 0/1 columns)

What to submit, and how it is marked

Submit **one Jupyter notebook** (`.ipynb`) that has been run from top to bottom, with all outputs visible. Give each scenario its own heading. Under each heading put the code, the output, and your written answer in a text cell. Write your answers in plain language, as if you were replying to the person who asked.

Marks per scenario	What earns them
2	The code runs and the numbers are correct.
1	The table or chart is readable: a title, labelled axes, and values a manager can read without the code.
1	The written answer responds to the business question and is supported by your numbers.

25 scenarios × 4 marks = **100 marks**. A scenario with correct code but no written answer earns at most 3 of 4. An answer that is not supported by the numbers in your own notebook earns no marks for the written part.